

Mockito Hands-On Exercises

Exercise 1: Mocking and Stubbing

MyServiceTest.java

```
package com.example; import static org.junit.jupiter.api.Assertions.assertEquals; import static
org.mockito.Mockito.when; import org.junit.jupiter.api.Test; import org.mockito.Mockito; public
class MyServiceTest { @Test void testExternalApi() { ExternalApi mockApi =
Mockito.mock(ExternalApi.class); when(mockApi.getData()).thenReturn("Mock Data"); MyService
service = new MyService(mockApi); String result = service.fetchData(); System.out.println("Fetched
Data: " + result); assertEquals("Mock Data", result); } }
```

Stubbing Console Output:

Console Log Output

```
Running testExternalApi...
Fetched Data: Mock Data
Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.812 s - in com.example.MyServiceTest
```

Exercise 2: Verifying Interactions

UserServiceTest.java

```
package com.example; import static org.mockito.Mockito.mock; import static
org.mockito.Mockito.verify; import org.junit.jupiter.api.Test; public class UserServiceTest {
@Test void testRegisterUser() { EmailService mockEmailService = mock(EmailService.class);
UserService userService = new UserService(mockEmailService);
userService.registerUser("sasank@example.com");
verify(mockEmailService).sendEmail("sasank@example.com"); } }
```

Interactions verification logs:

Console Log Output

```
Running testRegisterUser...
Registering user with email: sasank@example.com
Sending email to: sasank@example.com
Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.320 s - in com.example.UserServiceTest
```